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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 203 — Past Questions & Materials

by Renance

Course: MTH 203

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 200

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Paper 1 of 1 — MTH 203 Linear Algebra 1 2020 2021 First Semester

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
SECOND SEMESTER EXAMINATION 2020/2021

Course Code/Title: MTH 213

Course Title: Linear Algebra I

Instructions: Attempt any 4 questions. Time Allowed: 2 Hours.

Question 1

(a) Define the following subsets of Real numbers

(i) Natural numbers (ii) Whole numbers (iii) Integers (iv) Rational numbers

(4 marks)

(b) Write the vector W as a linearly combination of the vectors V_1, V_2 and V_3 .

(13.5 marks)

$$V_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, V_2 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, V_3 = \begin{pmatrix} -4 \\ 3 \\ 4 \end{pmatrix} \text{ and } W = \begin{pmatrix} 9 \\ 0 \\ -2 \end{pmatrix}.$$

Question 2

(a) Define a vector space and list all its axioms.

(6 marks)

(b) Determine if the vectors V_1, V_2 and V_3 are linearly independent, where

(11.5 marks)

$$V_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, V_2 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, V_3 = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix}$$

Question 3

Given the following matrices $A = \begin{bmatrix} 2 & -2 \\ 3 & 1 \\ -1 & 3 \end{bmatrix}$ $B = \begin{bmatrix} -4 & 6 & -5 \\ 5 & -3 & 2 \end{bmatrix}$ $C = \begin{bmatrix} -3 & -2 \\ 4 & 6 \\ 8 & -2 \end{bmatrix}$ $D = \begin{bmatrix} -2 & -2 & 4 \\ 6 & 1 & 3 \end{bmatrix}$.

Evaluate

(a) (i) AB (ii) $(BA)^2$

(b) (i) $(DC)^2$ (ii) CD

(9 marks)

(8.5 marks)

Question 4

(a) Find the determinant of the matrix $A = \begin{bmatrix} 2 & 5 & -4 \\ -3 & 6 & 1 \\ -5 & -2 & 3 \end{bmatrix}$

(5 marks)

(b) List 5 properties of determinants and justify each property with a solved example.

(12.5 marks)

Question 5

(a) Find the inverse of A (A^{-1}) given that $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 5 & 3 \\ -1 & 1 & 2 \end{bmatrix}$

(12.5 marks)

(b) Given $B = \begin{bmatrix} -4 & 6 & -5 \\ 5 & -3 & 2 \end{bmatrix}$ $C = \begin{bmatrix} -3 & -2 \\ 4 & 6 \\ 8 & -2 \end{bmatrix}$, find the trace of the transpose of the matrix BC . (5 marks)

Question 6

(a) Let $X = \mathbb{R}^2$ and $A = \begin{bmatrix} 2 & 5 \\ -2 & 3 \end{bmatrix}$. Define $T: X \rightarrow X$ by $Tz = Az$, where $z = \begin{bmatrix} x \\ y \end{bmatrix} \in X$. Show that T is a linear transformation and hence find the Kernel and Range of T

(12.5 marks)

(b) Let $A = (2, 3, 4)$ and $B = (8, -2, 5)$ be two vectors, find the cross (vector) product of A and B ($A \times B$)

(5 marks)



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